

A Dyadic Completion of the Thue–Morse Product

Lognormal moments, horizontal Dirichlet zero strings, and certified all-orders asymptotics

Research article prepared for Vladimir Reshetnikov

An extension of the ProveIt Thue–Morse atlas

4 September 2026

Abstract

The familiar Thue–Morse product $P(t) = \prod_{j \geq 0} (1 - e^{-2^j t})$ has a particularly useful completion. If $X = \sum_{j \geq 1} 2^{-j} U_j$, with independent uniform U_j on $[0, 1]$, and $\Phi(t) = \mathbb{E} e^{-tX}$, then $K(t) = P(t)\Phi(t)$ satisfies exactly $K(2t) = K(t)/t$. Its Mellin transform $C(s)$ therefore satisfies $C(s+1) = 2^{s+1}C(s)$. The resulting positive density has all integer moments of a lognormal distribution, but is not lognormal.

The central application concerns the entire continuation of $D(s, a) = \sum_{n \geq 0} (-1)^{s_2(n)} (n + a)^{-s}$, for $a > 0$. We prove a locally uniform all-orders scaling formula for $\Gamma(s - m)D(s - m, a)$. Its limiting function is $C(s)$, and every coefficient is an explicit polynomial-exponential multiple of $C(s)$. A nonconstant, entire, finite-order periodic factor of C must have nonreal zeros. Rouché’s theorem transfers each such zero s_0 to zeros of $D(s, a)$ near $s_0 - m$, with errors smaller than 2^{-Am} for every fixed $A > 0$. Thus every positive shift has infinitely many nonreal zeros in the left half-plane. For the half-shift, the error can be bounded on a Gaussian scale in m .

We also derive finite coefficient formulas for the derivatives at the real zeros. At the centered shift $a = 1/2$, their normalized values are samples of one completely monotone function. Its divergent expansion has an alternating remainder bounded by its first omitted term at every order. Explicit coefficient asymptotics give an optimized error of size $\exp[-(\log 2)m^2/2 + O(m)]$. The proofs, numerical checks, and a careful distinction between classical ingredients and research extensions are supplied in full.

Contents

1	Scope, provenance, and principal conclusions	3
1.1	What is being extended	3
1.2	Research status and attribution	3
1.3	The main theorem in one place	3
2	From binary cancellation to a completed product	4
2.1	The elementary finite product	4
2.2	The compact uniform series	4
2.3	The periodic factor is genuinely present	6
3	The entire Mellin transform and its periodic factor	6
3.1	Convergence and exact shifts	6
3.2	Fourier–Gaussian representation	7
3.3	Why a nonreal zero must exist	7
4	The Dirichlet family: continuation and real zeros	8
4.1	Entire continuation with a positive Mellin numerator	8
4.2	Entire growth	9

5	Exact rescaling and all-orders coefficient formulas	9
5.1	A bounded reciprocal transform	9
5.2	The rescaling identity	10
5.3	Nonrecursive finite formulas for the coefficients	11
5.4	All-orders slopes at the real zeros	11
6	Horizontal strings of nonreal Dirichlet zeros	12
6.1	Transferring a zero of the limiting transform	12
6.2	The historical zero question	12
6.3	What the theorem does not classify	13
7	Centering, complete monotonicity, and signed remainders	13
7.1	The centered reciprocal transform	13
7.2	One positive function contains every centered slope	14
7.3	Additional inequalities without asymptotic notation	15
8	Coefficient asymptotics and optimized error bounds	15
8.1	The first pole controls the coefficients	15
8.2	An explicit optimal-truncation estimate	16
9	Gaussian localization of the centered zero strings	17
10	A non-lognormal measure with exact lognormal moments	18
10.1	Moment equality and genuine distinctness	18
10.2	Size bias and the periodic degree of freedom	19
10.3	An exact mixture over geometric lattices	19
11	Exact moment algebra: determinants and orthogonal polynomials	20
12	Identical asymptotic series, different positive functions	21
13	Extension to sparse digit sequences in other integer bases	22
14	An explicit Fourier formula for the logarithmic profile	23
15	Numerical experiments and reproducibility	24
15.1	What was computed	24
15.2	Stable product evaluation	25
15.3	Why the Mellin quadrature is manageable	26
15.4	Reproduction commands	26
16	Relation to the atlas and a formalization route	26
16.1	Repository audit and limits of the novelty claim	26
16.2	A proof-oriented dependency chain	27
17	Further questions suggested by the proved results	27
A	Coefficient register and normalization checks	28
B	A direct nonrecursive formula for every asymptotic coefficient	29

1 Scope, provenance, and principal conclusions

1.1 What is being extended

The source atlas [1] already develops the binary recurrences, Prouhet cancellation, dyadic finite products, power moments, Fourier representations, rarefied sums, and the Mellin continuation of the Thue–Morse Dirichlet family. Repeating those chapters would not be a useful extension. This article instead connects three pieces of that framework: the lacunary Laplace product, the compactly supported uniform-convolution law underlying the Fabius function, and the Dirichlet continuation at large negative arguments.

The notation is independent of the repository’s macro files, so this source compiles without a checkout of the repository. We write

$$\varepsilon_n = (-1)^{s_2(n)}, \quad L = \log 2, \quad D(s, a) = \sum_{n \geq 0} \frac{\varepsilon_n}{(n + a)^s}. \quad (1.1)$$

Here $s_2(n)$ is the number of 1’s in the binary expansion of n . The shift a is always a positive real number unless explicitly stated otherwise. A prime on D means differentiation with respect to s . The Mellin transform uses the measure $t^{s-1} dt$.

1.2 Research status and attribution

Entireness of the Thue–Morse Dirichlet continuation is classical: see Allouche–Cohen [2], the general automatic-series theory [3], and the subsequent treatment [5]. The uniform series and its Fabius connection are also established background [9, 10, 1]. Lognormal moment indeterminacy, size-bias scaling, and geometric-lattice representations are classical [7]. None of those subjects is claimed as newly discovered here.

The principal research contribution of the present derivation is the *combined* completion–Mellin–zero mechanism, together with the coefficientwise factorization of the scaled Dirichlet expansion and the global signed remainder theorem at the half-shift. These are proved below, not merely suggested by experiments. Their mathematical validity and their bibliographical priority are separate questions. The literature search did not establish priority, and one relevant 2001 preprint by Alkauskas [6] could not be retrieved. Consequently “proved here” never means “proved here for the first time in the literature.” A more detailed audit is in section 16.

The zero theorem has a specific historical consequence. Section 3.2 of Allouche’s survey [4] asks whether the nontrivial zeros of $D(s, 1)$ consist only of $2\pi ik/L$. The theorem below supplies zeros with negative real part, and hence a negative answer to that exact question. This is a consequence of the proof, not a claim that the question has remained unresolved until the date of this article.

1.3 The main theorem in one place

Define

$$P(t) = \prod_{j \geq 0} (1 - e^{-2^j t}), \quad \Phi(t) = \prod_{j \geq 1} \frac{1 - e^{-t/2^j}}{t/2^j}, \quad K(t) = P(t)\Phi(t). \quad (1.2)$$

For $t > 0$ all three functions are positive. Put

$$C(s) = \int_0^\infty t^{s-1} K(t) dt, \quad C_0 = C(0), \quad Q(s) = \frac{C(s)}{C_0 2^{s(s+1)/2}}. \quad (1.3)$$

The powers of 2 always mean e^{Lz} , with no branch choice.

Theorem 1.1 (Completion and horizontal zero strings). *The integral $C(s)$ is entire; Q is entire, 1-periodic, positive on the real axis, nonconstant, and of order at most two. In particular, C has a nonreal zero.*

If s_0 is any zero of C of multiplicity d , then for each fixed $a > 0$ and all sufficiently large integers m , $D(\cdot, a)$ has a cluster of d zeros near $s_0 - m$. For every $A > 0$ these zeros lie in

$$|s - (s_0 - m)| < 2^{-Am} \quad (1.4)$$

for all sufficiently large m depending on A, a, s_0 . The clusters for different sufficiently large m are disjoint. For $a = 1/2$, the stronger bound

$$|s - (s_0 - m)| \leq \exp\left(-\frac{L}{2d}m^2 + O_{s_0}(m)\right) \quad (1.5)$$

holds. Every $a > 0$ therefore has infinitely many nonreal zeros in the left half-plane, including the unshifted-denominator case $a = 1$.

The ingredients and proof of the first bound occupy [sections 2 to 6](#). The quantitative refinement is proved in [section 9](#). The use of “cluster” allows multiple zeros of C ; simplicity is not assumed. No numerical zero enclosure is needed for the theorem.

2 From binary cancellation to a completed product

2.1 The elementary finite product

The binary identities

$$\varepsilon_{2n} = \varepsilon_n, \quad \varepsilon_{2n+1} = -\varepsilon_n \quad (2.1)$$

give, by expanding one factor for each binary digit,

$$\sum_{n=0}^{2^m-1} \varepsilon_n z^n = \prod_{j=0}^{m-1} (1 - z^{2^j}). \quad (2.2)$$

For $|z| < 1$, both sides tend to $\sum_{n \geq 0} \varepsilon_n z^n = \prod_{j \geq 0} (1 - z^{2^j})$. The products converge normally on compact subsets of the unit disk because $\sum_j |z|^{2^j} < \infty$ there. In particular,

$$P(t) = \sum_{n \geq 0} \varepsilon_n e^{-nt}, \quad P(2t) = \frac{P(t)}{1 - e^{-t}} \quad (t > 0). \quad (2.3)$$

The same product defines a holomorphic function for $\operatorname{Re} t > 0$.

For each integer $r \geq 0$, using the first r factors and bounding the others by one gives

$$0 < P(t) \leq 2^{r(r-1)/2} t^r. \quad (2.4)$$

Thus P is smaller than every fixed power at the origin. This cancellation is the analytic counterpart of the vanishing of the first m power moments in [equation \(2.2\)](#).

2.2 The compact uniform series

Let $U_1, U_2, \dots$ be independent random variables uniformly distributed on $[0, 1]$, and define

$$X = \sum_{j=1}^{\infty} 2^{-j} U_j. \quad (2.5)$$

The series converges surely, and $0 \leq X \leq 1$. Independence and bounded convergence give the entire Laplace transform

$$\Phi(z) = \mathbb{E}e^{-zX} = \prod_{j \geq 1} \frac{1 - e^{-z/2^j}}{z/2^j}. \quad (2.6)$$

Each factor is interpreted by continuity at $z = 0$. Normal convergence on compact sets follows either from its expansion $1 - z/2^{j+1} + O(4^{-j})$ or from boundedness of X . The distribution is symmetric about $1/2$, because replacing every U_j by $1 - U_j$ changes X to $1 - X$. Furthermore,

$$\mathbb{E}X = \frac{1}{2}, \quad \text{Var}(X) = \frac{1}{12} \sum_{j \geq 1} 4^{-j} = \frac{1}{36}. \quad (2.7)$$

The density of X , or a linearly rescaled version of it, is one of the standard uniform-convolution descriptions of the Fabius–Rvachev family. Using X avoids any ambiguity between conventions for those functions.

Separating the first factor in [equation \(2.6\)](#) gives

$$\Phi(2t) = \frac{1 - e^{-t}}{t} \Phi(t). \quad (2.8)$$

This factor is exactly the inverse of the nonmonomial factor in the scaling equation for P .

Theorem 2.1 (Exact dyadic completion). *For every $t > 0$,*

$$\boxed{K(2t) = \frac{K(t)}{t}}. \quad (2.9)$$

For every integer k , positive or negative,

$$K(2^k t) = 2^{-k(k-1)/2} t^{-k} K(t). \quad (2.10)$$

There is a positive, real-analytic, L -periodic function H such that

$$\boxed{K(t) = t^{1/2} \exp\left(-\frac{(\log t)^2}{2L}\right) H(\log t)}. \quad (2.11)$$

In particular, constants $0 < h_- \leq h_+ < \infty$ exist with

$$h_- t^{1/2} e^{-(\log t)^2/(2L)} \leq K(t) \leq h_+ t^{1/2} e^{-(\log t)^2/(2L)}. \quad (2.12)$$

Proof. Multiply [equation \(2.3\)](#) and [equation \(2.8\)](#) to obtain [equation \(2.9\)](#). Induction upward and downward proves [equation \(2.10\)](#). Set

$$H(v) = e^{v^2/(2L)-v/2} K(e^v).$$

Since $K(e^{v+L}) = e^{-v} K(e^v)$, direct substitution shows $H(v+L) = H(v)$. Positivity and continuity on $[0, L]$ give the two positive bounds. Holomorphy of $K(e^v)$ in $|\text{Im } v| < \pi/2$ gives real analyticity. The displayed representation and estimates are now identities and consequences of compactness, respectively. $\square$

2.3 The periodic factor is genuinely present

It is tempting to omit H , because its oscillation is numerically small. Doing so would destroy the zero theorem and would replace the actual moment measure by its lognormal twin.

Proposition 2.2 (Nonconstancy). *The function H in equation (2.11) is not constant.*

Proof. Suppose that $H \equiv c > 0$. Both $K(z)$ and

$$c z^{1/2} \exp\left(-\frac{(\operatorname{Log} z)^2}{2L}\right)$$

are holomorphic for $\operatorname{Re} z > 0$, where Log is the principal logarithm. Equality on the positive axis implies equality throughout that half-plane. Now take $z = x + 2\pi i$ with $x > 0$. Since 2^j is an integer,

$$P(x + 2\pi i) = P(x).$$

By equation (2.4), this tends to zero as $x \downarrow 0$. The entire function Φ remains bounded near $2\pi i$, so $K(x + 2\pi i) \rightarrow 0$. The proposed elementary expression, however, has a finite nonzero limit at $2\pi i$. This is a contradiction. $\square$

This proof uses only a boundary value and an identity theorem. In particular, it does not depend on numerical detection of a tiny Fourier coefficient.

3 The entire Mellin transform and its periodic factor

3.1 Convergence and exact shifts

Theorem 3.1 (Entire Mellin law). *The function C in equation (1.3) is entire and satisfies*

$$C(s + 1) = 2^{s+1}C(s), \tag{3.1}$$

$$C(s + j) = 2^{js+j(j+1)/2}C(s) \quad (j \in \mathbb{Z}). \tag{3.2}$$

For real σ , $C(\sigma) > 0$. Moreover,

$$|C(\sigma + i\tau)| \leq h_+ \sqrt{2\pi L} e^{L(\sigma+1/2)^2/2}. \tag{3.3}$$

Proof. With $t = e^v$, the integrand becomes

$$C(s) = \int_{-\infty}^{\infty} e^{-v^2/(2L)+(s+1/2)v} H(v) dv. \tag{3.4}$$

On every compact set of s , this integrand and every derivative in s are bounded by an integrable Gaussian times a polynomial in v . Differentiation under the integral proves entireness. Completing the square proves equation (3.3), and positivity is immediate. Changing variables $t = 2u$ in $C(s + 1)$ gives

$$C(s + 1) = 2^{s+1} \int_0^{\infty} u^s K(2u) du = 2^{s+1}C(s).$$

Iteration in either direction gives equation (3.2). $\square$

It follows at once that $Q(s + 1) = Q(s)$ and $Q(\sigma) > 0$ for real σ . Dividing equation (3.3) by the absolute value of $C_0 e^{L(s^2+s)/2}$ yields a useful stronger bound:

$$|Q(\sigma + i\tau)| \leq B_Q e^{L\tau^2/2}, \quad B_Q = \frac{h_+ \sqrt{2\pi L} e^{L/8}}{C_0}. \tag{3.5}$$

The constant does not depend on σ .

3.2 Fourier–Gaussian representation

Let

$$h_k = \frac{1}{L} \int_0^L H(v) e^{-2\pi i k v / L} dv, \quad H(v) = \sum_{k \in \mathbb{Z}} h_k e^{2\pi i k v / L}. \quad (3.6)$$

Analyticity in a strip implies exponential decay of h_k ; in particular, the series converges absolutely and uniformly on the real line.

Proposition 3.2 (Gaussian smoothing in Mellin space). *For every $s \in \mathbb{C}$,*

$$C(s) = \sqrt{2\pi L} e^{L(s+1/2)^2/2} \sum_{k \in \mathbb{Z}} h_k e^{-2\pi^2 k^2 / L} e^{2\pi i k (s+1/2)}. \quad (3.7)$$

The series on the right converges normally on the entire s -plane. The periodic entire function Q is nonconstant.

Proof. Insert [equation \(3.6\)](#) in [equation \(3.4\)](#). Absolute summability and the Gaussian bound permit interchange. For each k , the elementary Gaussian integral gives

$$\int_{\mathbb{R}} e^{-v^2/(2L) + (s+1/2+2\pi i k/L)v} dv = \sqrt{2\pi L} e^{L(s+1/2+2\pi i k/L)^2/2}.$$

Expanding the square proves the formula. The quadratic decay in k proves normal convergence for all complex s .

By [proposition 2.2](#), some h_k with $k \neq 0$ is nonzero. Multiplication by $e^{-2\pi^2 k^2 / L}$ never annihilates a coefficient. Uniqueness of Fourier coefficients on the real period therefore shows that the sum in [equation \(3.7\)](#), and hence Q , is nonconstant. $\square$

The distinction between H and Q is important. The first harmonic of H is small, but the corresponding harmonic of Q is further multiplied by $e^{-2\pi^2/L}$. On the real axis Q is consequently extraordinarily close to a constant. At complex arguments with a moderate imaginary part, the factor $e^{2\pi i k s}$ can undo this suppression. That is where the nonreal zeros become visible.

3.3 Why a nonreal zero must exist

We record the elementary finite-order fact needed for the main argument rather than assume a zero-location conjecture.

Lemma 3.3 (A zero-free periodic function of finite order). *Let F be an entire, 1-periodic function satisfying $|F(z)| \leq e^{A+B|z|^2}$ for some constants. If F is positive on the real axis and has no zeros, then F is constant.*

Proof. Since $\mathbb{C}$ is simply connected and F has no zeros, there is an entire function g with $F = e^g$. The growth hypothesis bounds $\operatorname{Re} g(z)$ above by $A + B|z|^2$. The Borel–Carathéodory inequality on concentric disks of radii R and $R/2$ therefore gives

$$\max_{|z| \leq R/2} |g(z) - g(0)| = O(R^2).$$

For completeness, the form used here follows by applying the maximum principle and Schwarz’s lemma to $w/(2M-w)$, where $w = g - g(0)$ and $M = \max_{|z| \leq R} \operatorname{Re} w$; it gives $\max_{|z| \leq R/2} |w| \leq 2M$ when $M > 0$. Cauchy’s estimates now show that every coefficient of degree greater than two in the Taylor series of g vanishes. Thus $g(z) = az^2 + bz + c$.

Periodicity implies that $g(z+1) - g(z)$ takes values in the discrete set $2\pi i \mathbb{Z}$. It is continuous, so it is constant. Therefore $a = 0$ and $b = 2\pi i k$ for an integer k . Since $e^{c+2\pi i k x}$ is positive for every real x , $k = 0$. Thus F is constant. $\square$

Corollary 3.4 (Nonreal zeros of the completion). *The functions Q and C have nonreal zeros. Their zero sets agree, are invariant under complex conjugation and translation by integers, and contain no real point.*

Proof. The Gaussian factor relating Q and C never vanishes. Apply lemma 3.3 to equation (3.5) and use nonconstancy. Positivity excludes real zeros. Reality of K gives $C(\bar{s}) = \overline{C(s)}$, and equation (3.2) gives the translation invariance. $\square$

4 The Dirichlet family: continuation and real zeros

4.1 Entire continuation with a positive Mellin numerator

The partial sums of ε_n have absolute value at most one:

$$\sum_{n < 2N} \varepsilon_n = 0, \quad \sum_{n < 2N+1} \varepsilon_n = \varepsilon_N.$$

Summation by parts therefore gives locally uniform convergence of equation (1.1) in $\operatorname{Re} s > 0$. Define

$$M(s, a) = \int_0^\infty t^{s-1} e^{-at} P(t) dt. \quad (4.1)$$

The flatness bound equation (2.4) handles every compact set of s at the origin, and e^{-at} handles infinity. Thus M is entire. In $\operatorname{Re} s > 1$, absolute convergence permits termwise integration, giving

$$D(s, a) = \frac{M(s, a)}{\Gamma(s)}. \quad (4.2)$$

Because $1/\Gamma$ is entire, this is the entire continuation. This recovers a classical result in the normalization needed here.

Proposition 4.1 (All real zeros, their signs, and their slopes). *For every $a > 0$, the real zeros of $D(s, a)$ are exactly $0, -1, -2, \dots$, and every one is simple. Writing*

$$Z_m(a) = \frac{(-1)^m}{m!} D'(-m, a), \quad m \geq 0, \quad (4.3)$$

one has

$$Z_m(a) = M(-m, a) > 0. \quad (4.4)$$

For every $k \geq 0$,

$$(-1)^k \frac{\partial^k}{\partial a^k} Z_m(a) = M(k - m, a) > 0. \quad (4.5)$$

Proof. For every real s , the integrand defining $M(s, a)$ is strictly positive. The only real zeros of $1/\Gamma(s)$ are the simple zeros at $s = -m$. The residue formula $\operatorname{Res}_{s=-m} \Gamma(s) = (-1)^m/m!$ gives

$$\frac{1}{\Gamma(s)} = (-1)^m m! (s + m) + O((s + m)^2).$$

Multiply by $M(s, a)$ to obtain equation (4.4) and simplicity. Differentiation in a is justified by the same endpoint bounds and introduces $(-t)^k$, proving equation (4.5). $\square$

These observations also show that the sign of $D(s, a)$ between successive real zeros is exactly the sign of $1/\Gamma(s)$. The sequence of slopes has a separate moment positivity property.

Corollary 4.2 (Strict moment inequalities for zero derivatives). *For each fixed $a > 0$, every finite Hankel matrix $[Z_{i+j}(a)]_{i,j=0}^r$ is positive definite. In particular,*

$$Z_m(a)Z_{m+2}(a) > Z_{m+1}(a)^2 \quad (m \geq 0). \quad (4.6)$$

Also $Z'_m(a) = -Z_{m-1}(a)$ for $m \geq 1$.

Proof. Substitute $u = 1/t$ in [equation \(4.4\)](#):

$$Z_m(a) = \int_0^\infty u^m e^{-a/u} P(1/u) \frac{du}{u}.$$

The measure is positive and has support all of $(0, \infty)$. The integral of the square of a nonzero polynomial is therefore strictly positive. This proves the matrix statement. Strict Cauchy–Schwarz applied to $u^{m/2}$ and $u^{m/2+1}$ proves the inequality. The derivative identity follows from [equation \(4.5\)](#). $\square$

4.2 Entire growth

Proposition 4.3 (Quadratic growth type). *For each fixed $a > 0$, both $M(\cdot, a)$ and $D(\cdot, a)$ have entire order two and type $L/2$, in the convention*

$$\limsup_{r \rightarrow \infty} \frac{\log \max_{|s|=r} |F(s)|}{r^2} = \frac{L}{2}.$$

Proof. For $0 < t \leq 1$, $\Phi(t) \geq e^{-t} \geq e^{-1}$ since $X \leq 1$. Thus [equation \(2.12\)](#) bounds the small- t part of M by a constant times a Gaussian integral in $\log t$, at most $\exp(Lr^2/2 + O(r))$ when $|s| \leq r$. For $t \geq 1$, $P(t) \leq 1$ and

$$\int_1^\infty t^{\operatorname{Re} s - 1} e^{-at} dt \leq \int_0^\infty t^r e^{-at} dt = a^{-r-1} \Gamma(r+1).$$

Its logarithm is $O_a(r \log r)$. These estimates give the required upper type for M . The standard reciprocal-Gamma product gives $\log \max_{|s| \leq r} |1/\Gamma(s)| = O(r \log r)$, so the same upper type holds for D .

For the lower bound, take $s = -r$. In the logarithmic integral for M , restrict v to a fixed-length interval around $v = L(1/2 - r)$. On this interval $e^{-ae^v}/\Phi(e^v)$ tends uniformly to one, and $H(v) \geq h_-$. Completing the square yields

$$M(-r, a) \geq c_a \exp\left(\frac{L}{2}(r - 1/2)^2\right)$$

for all sufficiently large r . This proves the lower type for M . For D , use $r = n + 1/2$. The reflection formula gives $|1/\Gamma(-r)| = \Gamma(1+r)/\pi$, which does not reduce the quadratic lower bound. The asserted order and type follow. $\square$

5 Exact rescaling and all-orders coefficient formulas

5.1 A bounded reciprocal transform

Define the analytic germ

$$B_a(z) = \frac{e^{-az}}{\Phi(z)} = \sum_{j \geq 0} \beta_j(a) z^j. \quad (5.1)$$

The nearest zeros of Φ are at $z = \pm 4\pi i$, so the radius of this Taylor series is exactly 4π . To verify that no extra zero is hidden inside that disk, note that each factor in the normally convergent product for Φ is nonzero there and that the sum of its deviations from one converges normally. At $4\pi i$ the first factor has a zero, while the numerator in [equation \(5.1\)](#) is nonzero.

Lemma 5.1 (Global Taylor remainder on the positive axis). *For each $a > 0$ and integer $N \geq 0$, there is a finite constant $B_{a,N}$ such that*

$$\left| B_a(t) - \sum_{j=0}^{N-1} \beta_j(a) t^j \right| \leq B_{a,N} t^N \quad (t \geq 0). \quad (5.2)$$

The constants can be chosen uniformly when a belongs to a compact subinterval of $(0, \infty)$.

Proof. First, $\mathbb{P}(X \leq \delta) > 0$ for every $\delta > 0$. Choose J so that $\sum_{j>J} 2^{-j} \leq \delta/2$, and restrict the finitely many U_j , $j \leq J$, to a sufficiently small interval. This event has positive probability and forces $X \leq \delta$. Hence, for $0 < \delta < a$,

$$\Phi(t) \geq \mathbb{P}(X \leq \delta) e^{-\delta t}, \quad 0 < B_a(t) \leq \frac{e^{-(a-\delta)t}}{\mathbb{P}(X \leq \delta)}.$$

Thus B_a is bounded and tends exponentially to zero on the positive axis. Near zero, analyticity bounds the remainder divided by t^N . On a compact interval away from zero the quotient is continuous; at infinity boundedness of B_a and the degree of the subtracted polynomial bound the quotient. Taking the supremum proves the claim. For a compact shift interval, fix δ smaller than its left endpoint and use compactness for the local Taylor bounds. $\square$

This lemma does not assert convergence of the Taylor series for all positive t . It states a separate global inequality for each finite truncation. That distinction is essential when the integral samples arbitrarily large t .

5.2 The rescaling identity

Theorem 5.2 (Exact renormalization). *For $m \geq 0$ an integer, put $h = 2^{-m}$ and*

$$F_{m,a}(s) = 2^{-m(m-1)/2+ms} M(s-m, a). \quad (5.3)$$

Then, for every $s \in \mathbb{C}$,

$$F_{m,a}(s) = \int_0^\infty u^{s-1} K(u) B_a(hu) \, du. \quad (5.4)$$

For every compact $\mathcal{K} \subset \mathbb{C}$ and fixed integer $N \geq 1$,

$$F_{m,a}(s) = C(s) \sum_{j=0}^{N-1} \beta_j(a) 2^{js+j(j+1)/2} h^j + O_{\mathcal{K},a,N}(h^N). \quad (5.5)$$

The remainder is locally uniform in s and in positive a .

Proof. Write $e^{-at} P(t) = K(t) B_a(t)$ in [equation \(4.1\)](#), and set $t = hu$. The negative-index case of [equation \(2.10\)](#) says

$$K(hu) = 2^{-m(m+1)/2} u^m K(u).$$

Consequently

$$M(s-m, a) = 2^{m(m-1)/2-ms} \int_0^\infty u^{s-1} K(u) B_a(hu) \, du,$$

which is the exact identity.

Insert the finite Taylor expansion from [lemma 5.1](#). The j th integral is $\beta_j(a) h^j C(s+j)$; use [equation \(3.2\)](#). If $s = \sigma + i\tau$, the absolute value of the remainder integral is at most

$$B_{a,N} h^N C(\sigma + N). \quad (5.6)$$

Since C is continuous on the real line, this is uniform for s in a compact set. The uniform shift statement follows from the last part of [lemma 5.1](#). $\square$

The factorization in [equation \(5.5\)](#) is stronger than ordinary asymptotic convergence. Every coefficient vanishes at *every* zero of C . This is precisely why the zero displacement is smaller than any fixed geometric scale.

5.3 Nonrecursive finite formulas for the coefficients

The elementary Bernoulli expansion

$$\log \frac{1 - e^{-z}}{z} = -\frac{z}{2} + \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!} z^{2r}$$

yields, after summing over dyadic scales,

$$\log B_a(z) = \left(\frac{1}{2} - a\right) z - \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!(2^{2r} - 1)} z^{2r}. \quad (5.7)$$

This converges for $|z| < 4\pi$. Let

$$\lambda_1 = \frac{1}{2} - a, \quad \lambda_{2r} = -\frac{B_{2r}}{2r(2r)!(2^{2r} - 1)}, \quad \lambda_{2r+1} = 0 \quad (r \geq 1). \quad (5.8)$$

Then the ordinary coefficient, with no auxiliary recurrence, is

$$\beta_j(a) = \sum_{\substack{n_1, \dots, n_j \geq 0 \\ n_1 + 2n_2 + \dots + jn_j = j}} \prod_{r=1}^j \frac{\lambda_r^{n_r}}{n_r!}. \quad (5.9)$$

It follows by multiplying the exponential series $\prod_r \exp(\lambda_r z^r)$. Equivalently, $j! \beta_j$ is a complete exponential Bell polynomial evaluated at $r! \lambda_r$. The derivative relation $\partial_a \beta_j(a) = -\beta_{j-1}(a)$ for $j \geq 1$ is immediate from [equation \(5.1\)](#).

With $b = 1/2 - a$, the first coefficients are

$$\begin{aligned} \beta_0 &= 1, & \beta_1 &= b, & \beta_2 &= \frac{b^2}{2} - \frac{1}{72}, \\ \beta_3 &= \frac{b^3}{6} - \frac{b}{72}, & \beta_4 &= \frac{b^4}{24} - \frac{b^2}{144} + \frac{31}{259200}. \end{aligned} \quad (5.10)$$

5.4 All-orders slopes at the real zeros

Taking $s = 0$ in [equation \(5.5\)](#) gives the promised derivative formula.

Corollary 5.3 (Zero-derivative expansion). *For every fixed $a > 0$ and $N \geq 1$,*

$$\frac{(-1)^m D'(-m, a)}{m!} = C_0 2^{m(m-1)/2} \left\{ \sum_{j=0}^{N-1} 2^{j(j+1)/2} \beta_j(a) 2^{-mj} + O_{a,N}(2^{-mN}) \right\}. \quad (5.11)$$

In particular,

$$\begin{aligned} \frac{Z_m(a)}{C_0 2^{m(m-1)/2}} &= 1 + 2b 2^{-m} + \left(4b^2 - \frac{1}{9}\right) 2^{-2m} \\ &\quad + \left(\frac{32}{3}b^3 - \frac{8}{9}b\right) 2^{-3m} + O_a(2^{-4m}). \end{aligned} \quad (5.12)$$

The half-shift is the unique value of a for which all the odd coefficients vanish. It is not merely a convenient numerical choice; it corresponds exactly to centering the compact random variable X .

6 Horizontal strings of nonreal Dirichlet zeros

6.1 Transferring a zero of the limiting transform

Theorem 6.1 (Supergeometric localization). *Let s_0 be a zero of C of multiplicity d . Fix $a > 0$. For every $A > 0$, and every sufficiently large integer m , $D(\cdot, a)$ has exactly d zeros, counted with multiplicity, in*

$$|s - (s_0 - m)| < 2^{-Am}. \quad (6.1)$$

These zeros are nonreal. In particular, there are infinitely many nonreal zeros with negative real part.

Proof. Choose a fixed disk $\mathcal{K}$ about s_0 containing no other zero of C and not meeting the real axis. In a sufficiently small disk, write

$$C(s) = (s - s_0)^d G(s), \quad |G(s)| \geq g_0 > 0.$$

Let $h = 2^{-m}$ and choose an integer $N > Ad$. Define

$$S_N(s, h) = \sum_{j=0}^{N-1} \beta_j(a) 2^{js+j(j+1)/2} h^j.$$

For this fixed N , $S_N(s, h) \rightarrow 1$ uniformly on $\mathcal{K}$. It is consequently nonzero there and has modulus at least $1/2$ for large m . By [theorem 5.2](#),

$$F_{m,a}(s) = C(s)S_N(s, h) + R_{m,N}(s), \quad \sup_{s \in \mathcal{K}} |R_{m,N}(s)| \leq c_N h^N.$$

On the circle $|s - s_0| = h^A$,

$$|C(s)S_N(s, h)| \geq \frac{g_0}{2} h^{Ad}.$$

Because $N > Ad$, this is strictly larger than $c_N h^N$ for large m . Rouché’s theorem gives exactly d zeros of $F_{m,a}$ inside the circle. The nonzero prefactor in [equation \(5.3\)](#) shows that the corresponding points shifted by $-m$ are zeros of $M(\cdot, a)$. They remain nonreal, so $1/\Gamma$ is nonzero there and they are also zeros of $D(\cdot, a)$, with the same multiplicities.

Finally, [corollary 3.4](#) supplies at least one such s_0 . The centers $s_0 - m$ are separated by distance one, while the radii tend to zero. The resulting disks are disjoint and lie in the left half-plane for all sufficiently large m . $\square$

The threshold in this theorem depends on A . The assertion is not that one fixed numerical error bound works simultaneously for all A and all m . Rather, every fixed geometric scale is eventually too coarse to resolve the displacement.

A standard $N = 1$ application of Rouché would only give convergence to a translated zero. The all-orders factorization proves much more: there is no nonzero formal power series in h describing the movement of the zero. Its displacement is a genuinely beyond-all-orders effect.

6.2 The historical zero question

For $a = 1$, the series is exactly

$$D(s, 1) = \sum_{n \geq 0} \frac{(-1)^{s_2(n)}}{(n+1)^s},$$

the series in Section 3.2 of [4]. Theorem 6.1 gives infinitely many zeros with negative real part and nonzero imaginary part. They are neither nonpositive integers nor points $2\pi i k/L$. Thus the proposed exhaustion of the nontrivial zero set by that imaginary lattice is false.

The proof does not rely on the known imaginary-axis zeros, on their multiplicities, or on any unproved property of $\zeta(s)$. Its main inputs are a positive product, an entire periodic function of finite order, and Rouché’s theorem. The particular zero displayed numerically later is an illustration; existence is already forced by [lemma 3.3](#).

For comparisons between the centered and the original family, one also has the exact elementary identity

$$\boxed{D(s, 1/2) = (1 + 2^s)D(s, 1).} \quad (6.2)$$

Indeed, separating even and odd indices in $D(s, 1)$ gives $D(s, 1) = 2^{-s}D(s, 1/2) - 2^{-s}D(s, 1)$ in the region of absolute convergence, and entireness extends the identity to all s . The multiplier has zeros only on the imaginary axis. Hence every nonreal zero off that axis in the half-shift family is also a zero of $D(s, 1)$, and conversely. This provides an additional normalization check on the numerical examples.

6.3 What the theorem does not classify

The theorem does not enumerate all zeros of Q in a fundamental strip, prove that they are simple, or exclude other families of zeros of D . It also does not assert that every zero in a left half-plane lies in one of the strings when the imaginary part is allowed to grow with m . The asymptotic theorem is locally uniform in the translated variable s , so it controls bounded-height windows. Those distinctions are useful when formulating further questions rather than overstating the result.

7 Centering, complete monotonicity, and signed remainders

7.1 The centered reciprocal transform

At $a = 1/2$, symmetry of X gives

$$\Psi(z) := B_{1/2}(z) = \prod_{j \geq 1} \frac{z/2^{j+1}}{\sinh(z/2^{j+1})}. \quad (7.1)$$

This is an even meromorphic function. We next represent its restriction to the positive axis by a positive Laplace transform. That positivity will certify the sign and size of every finite remainder.

Lemma 7.1 (An exponential-sum representation). *Let $E_{j,n}$, $j, n \geq 1$, be independent exponential random variables of mean one, and put*

$$W = \sum_{j,n \geq 1} \frac{E_{j,n}}{\pi^2 2^{2j+2} n^2}. \quad (7.2)$$

Then W is finite and positive almost surely, $\mathbb{E}W = 1/72$, and

$$\boxed{\Psi(\sqrt{x}) = \mathbb{E}e^{-xW} \quad (x \geq 0).} \quad (7.3)$$

All positive moments of W are finite. If

$$c_r = \frac{\mathbb{E}W^r}{r!}, \quad (7.4)$$

then $c_r > 0$, and $\Psi(z) = \sum_{r \geq 0} (-1)^r c_r z^{2r}$ for $|z| < 4\pi$.

Proof. The sum of the expectations in [equation \(7.2\)](#) is

$$\frac{\zeta(2)}{\pi^2} \sum_{j \geq 1} 2^{-2j-2} = \frac{1}{6} \cdot \frac{1}{12} = \frac{1}{72}.$$

A nonnegative random sum with finite expectation is finite almost surely. Euler’s product $\sinh z/z = \prod_{n \geq 1} (1 + z^2/(\pi^2 n^2))$ now gives

$$\Psi(\sqrt{x}) = \prod_{j,n \geq 1} \left(1 + \frac{x}{\pi^2 2^{2j+2} n^2}\right)^{-1}.$$

Each factor is the Laplace transform of the corresponding term in [equation \(7.2\)](#). First multiply finitely many factors, then pass to the almost sure limit by dominated convergence.

The least denominator in [equation \(7.2\)](#) is $16\pi^2$. For $0 < y < 16\pi^2$, the product $\prod_{j,n} (1 - y/(\pi^2 2^{2j+2} n^2))^{-1}$ converges, because the sum of the inverse denominators converges and only finitely many can be close to y . Thus $\mathbb{E}e^{yW} < \infty$, so every moment is finite. Expanding this moment generating function gives the claimed Taylor coefficients and their positivity. $\square$

The coefficients also have a finite positive expression. Define

$$\gamma_r = \frac{|B_{2r}|}{2r(2r)!(2^{2r} - 1)}. \quad (7.5)$$

Then

$$c_r = \sum_{\substack{n_1, \dots, n_r \geq 0 \\ n_1 + 2n_2 + \dots + rn_r = r}} \prod_{k=1}^r \frac{\gamma_k^{n_k}}{n_k!}. \quad (7.6)$$

Indeed, substituting $z = i\sqrt{y}$ in [equation \(5.7\)](#) gives $\log \mathbb{E}e^{yW} = \sum_{r \geq 1} \gamma_r y^r$. Thus neither the definition of the coefficients nor their positivity requires solving a recurrence.

7.2 One positive function contains every centered slope

Introduce the probability measure

$$\mu(dt) = \frac{K(t)}{C_0} \frac{dt}{t}, \quad t > 0, \quad (7.7)$$

and let T have this distribution, independently of W . Define

$$\mathcal{H}(x) = \mathbb{E}e^{-xT^2W} \quad (x \geq 0), \quad A_r = 2^{2r^2+r} c_r. \quad (7.8)$$

The exact integer moments of T follow immediately from [equation \(3.2\)](#):

$$\mathbb{E}T^n = \frac{C(n)}{C_0} = 2^{n(n+1)/2} \quad (n \in \mathbb{Z}). \quad (7.9)$$

These moments will be developed further in [section 10](#).

Theorem 7.2 (Exact centered interpolation and global remainder). *The function $\mathcal{H}$ is completely monotone on $[0, \infty)$, with finite right derivatives of every order at zero, and*

$$\frac{Z_m(1/2)}{C_0 2^{m(m-1)/2}} = \mathcal{H}(4^{-m}) \quad (m \geq 0). \quad (7.10)$$

For every integer $N \geq 0$ and every $x \geq 0$,

$$0 \leq (-1)^N \left[\mathcal{H}(x) - \sum_{r=0}^{N-1} (-1)^r A_r x^r \right] \leq A_N x^N. \quad (7.11)$$

For $x > 0$, the lower inequality is strict; the upper inequality is strict when $N \geq 1$. In particular, the normalized centered slopes are enclosed between successive even and odd truncations of the same expansion.

Proof. The Laplace representation makes complete monotonicity immediate:

$$(-1)^r \mathcal{H}^{(r)}(x) = \mathbb{E}[(T^2 W)^r e^{-xT^2 W}] > 0.$$

At zero the expectation is finite by [equation \(7.9\)](#) and [lemma 7.1](#); dominated convergence justifies the right derivatives. Furthermore,

$$\frac{\mathbb{E}(T^2 W)^r}{r!} = \mathbb{E} T^{2r} c_r = 2^{2r^2+r} c_r = A_r.$$

Taking $s = 0$, $a = 1/2$ in [equation \(5.4\)](#), dividing by C_0 , and using [equation \(7.3\)](#) gives [equation \(7.10\)](#).

For $y \geq 0$ and $N \geq 1$, the integral Taylor remainder is

$$e^{-y} - \sum_{r=0}^{N-1} \frac{(-y)^r}{r!} = \frac{(-1)^N}{(N-1)!} \int_0^y e^{-v} (y-v)^{N-1} dv.$$

Its signed value is positive for $y > 0$ and at most $y^N/N!$, strictly less when $y > 0$. Substitute $y = xT^2 W$ and take expectations. For $N = 0$, the statement is simply $0 < \mathcal{H}(x) \leq 1$. $\square$

The first terms are

$$\begin{aligned} \mathcal{H}(x) \sim 1 - \frac{x}{9} + \frac{248}{2025}x^2 - \frac{4806656}{2679075}x^3 \\ + \frac{3993751257088}{10247461875}x^4 - \dots \end{aligned} \quad (7.12)$$

For example, for every $m \geq 0$,

$$1 - \frac{4^{-m}}{9} < \frac{Z_m(1/2)}{C_0 2^{m(m-1)/2}} < 1 - \frac{4^{-m}}{9} + \frac{248}{2025}16^{-m}. \quad (7.13)$$

The inequalities are useful even though the full infinite series will be shown to diverge at every nonzero x .

7.3 Additional inequalities without asymptotic notation

Since $Y = T^2 W$ is positive and $\mathbb{E}Y = A_1 = 1/9$, Jensen's inequality for the convex function $y \mapsto e^{-xy}$ gives

$$e^{-x/9} \leq \mathcal{H}(x) \leq 1 \quad (x \geq 0). \quad (7.14)$$

For $x > 0$ the first inequality is strict because Y is not constant. The function $\mathcal{H}$ is also strictly log-convex:

$$\mathcal{H}(x)\mathcal{H}''(x) - \mathcal{H}'(x)^2 > 0. \quad (7.15)$$

To see this, apply strict Cauchy–Schwarz to 1 and Y under the positive measure $e^{-xY} d\mathbb{P}$. Consequently the quantity $-\mathcal{H}'(x)/\mathcal{H}(x)$ decreases strictly with x and lies between zero and $1/9$. These are genuine global constraints on the centered slope interpolant, not only statements as $m \rightarrow \infty$.

8 Coefficient asymptotics and optimized error bounds

8.1 The first pole controls the coefficients

Set

$$R = 16\pi^2, \quad B = 2 \prod_{k \geq 1} \frac{\pi/2^k}{\sin(\pi/2^k)}. \quad (8.1)$$

The product for B converges, because its logarithmic terms are $O(4^{-k})$.

Theorem 8.1 (Sharp leading coefficient growth). *The positive coefficients c_r satisfy*

$$c_r R^r \nearrow B, \quad c_r \sim B R^{-r}. \quad (8.2)$$

Consequently,

$$A_r \sim B \frac{2^{2r^2+r}}{(16\pi^2)^r}, \quad 0 < A_r \leq B \frac{2^{2r^2+r}}{(16\pi^2)^r}. \quad (8.3)$$

The series $\sum_{r \geq 0} (-1)^r A_r x^r$ has radius of convergence zero.

Proof. Write the moment generating function from lemma 7.1 as

$$\sum_{r \geq 0} c_r w^r = \prod_{j, n \geq 1} \left(1 - \frac{w}{\pi^2 2^{2j+2} n^2}\right)^{-1} = \frac{G(w)}{1 - w/R},$$

where the factor with $j = n = 1$ has been removed to define G . Every Taylor coefficient g_r of G is nonnegative. All its remaining pole parameters are at least $4R$, so G is analytic on $|w| < 4R$, and

$$c_r R^r = \sum_{k=0}^r g_k R^k \nearrow G(R).$$

The increase is strict because G has positive coefficients of positive degree. To evaluate $G(R)$, put $z = i\sqrt{w}$ in equation (7.1). The first factor satisfies

$$\lim_{w \rightarrow R} (1 - w/R) \frac{\sqrt{w}/4}{\sin(\sqrt{w}/4)} = 2.$$

The remaining factors tend to those in equation (8.1), so $G(R) = B$. This proves both the asymptotic and the upper bound. Finally, $A_r^{1/r} \sim 2^{2r+1}/R \rightarrow \infty$, proving zero radius. $\square$

Thus equation (7.12) is an all-orders asymptotic expansion, not a convergent power series. Increasing the number of terms helps only up to a truncation order that depends on x .

8.2 An explicit optimal-truncation estimate

Let $\ell = \log(1/x)$, and suppose $x > 0$ is sufficiently small that the nearest integer to the following expression is nonnegative:

$$N_*(x) = \frac{\ell + \log R - L}{4L}. \quad (8.4)$$

Choose any nearest nonnegative integer N to $N_*(x)$.

Corollary 8.2 (Certified logarithmic-Gaussian error). *For this truncation,*

$$\left| \mathcal{H}(x) - \sum_{r=0}^{N-1} (-1)^r A_r x^r \right| \leq B \exp \left\{ \frac{L}{2} - \frac{[\log(1/x) + \log(8\pi^2)]^2}{8L} \right\}. \quad (8.5)$$

At the dyadic samples $x = 4^{-m}$, this is a relative error bound for equation (7.10) of size $\exp[-Lm^2/2 + O(m)]$.

Proof. By equations (7.11) and (8.3), the error is at most

$$B \exp\{2LN^2 + (L - \log R - \ell)N\}.$$

Completing the square gives the minimum at equation (8.4), with value $-(\ell + \log R - L)^2/(8L)$ in the exponent. Rounding to a nearest integer increases the quadratic by at most $2L(1/2)^2 = L/2$. Since $\log R - L = \log(8\pi^2)$, the result follows. $\square$

This is a rigorous upper bound, including its explicit constant. It is not an assertion that the actual error has that leading asymptotic size. The distinction matters, especially at complex zeros where extra cancellation occurs.

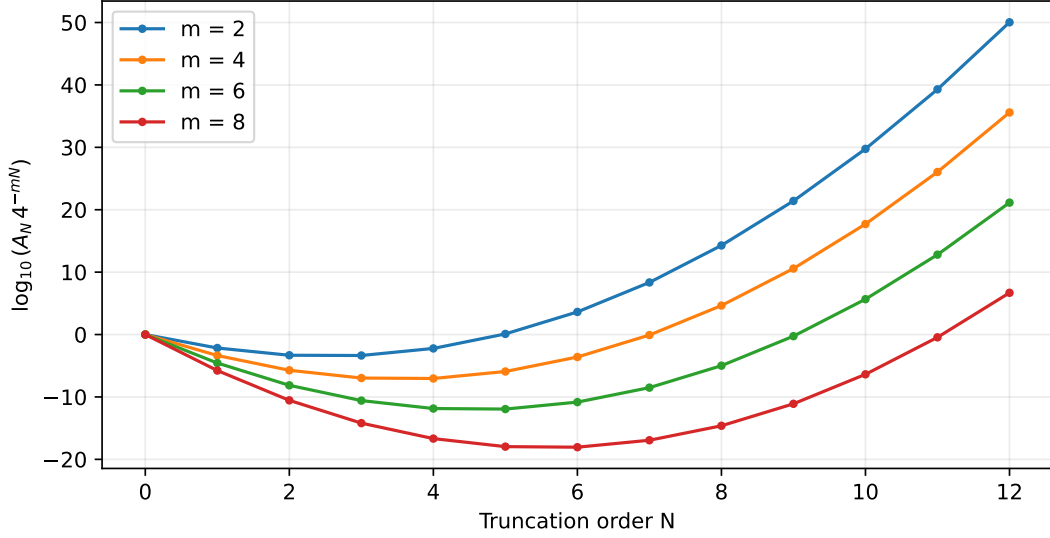


Figure 1: The proved first-omitted-term bounds $A_N 4^{-mN}$. Each curve first decreases and then increases. These are values of explicit bounds, not measured numerical errors.

9 Gaussian localization of the centered zero strings

The alternating inequality also controls complex translated arguments, although the complex remainder no longer has a sign. This is the quantitative refinement asserted in [theorem 1.1](#).

Proposition 9.1 (Complex centered remainder). *Let $h = 2^{-m}$, $s = \sigma + i\tau$, and $N \geq 0$. Then*

$$F_{m,1/2}(s) = C(s) \sum_{r=0}^{N-1} (-1)^r c_r 2^{2rs+2r^2+r} h^{2r} + R_{m,N}(s), \quad (9.1)$$

$$|R_{m,N}(s)| \leq c_N h^{2N} C(\sigma + 2N) = c_N h^{2N} 2^{2N\sigma+2N^2+N} C(\sigma). \quad (9.2)$$

For s in any fixed compact set, an integer $N = N(m) = m/2 + O(1)$ can be chosen so that

$$\sup_s |R_{m,N(m)}(s)| \leq \exp[-Lm^2/2 + O(m)]. \quad (9.3)$$

Proof. Use the Taylor remainder of $\Psi(t) = \mathbb{E}e^{-t^2 W}$ for real $t \geq 0$: its absolute value after N even terms is at most $c_N t^{2N}$. Insert it into [equation \(5.4\)](#); taking absolute values replaces u^{s-1} by $u^{\sigma-1}$. This proves the first inequality. The Mellin shift identity proves the equality.

On a compact set let $\sigma \leq b$, and bound $C(\sigma)$ by a constant. Using $c_N \leq BR^{-N}$, the logarithm of the bound is at most

$$2LN^2 + \{L(1 + 2b - 2m) - \log R\}N + O(1).$$

Its minimizing integer is $m/2 + O(1)$; substitution gives [equation \(9.3\)](#). $\square$

Theorem 9.2 (Centered Gaussian zero clusters). *If s_0 is a zero of C of multiplicity d , then the d zeros of $D(\cdot, 1/2)$ in the cluster approaching $s_0 - m$ satisfy*

$$|s - (s_0 - m)| \leq \exp[-Lm^2/(2d) + O_{s_0}(m)]. \quad (9.4)$$

The same conclusion holds for the corresponding off-axis zeros of $D(\cdot, 1)$.

Proof. Choose a small fixed disk $\mathcal{K}$ about s_0 as in [theorem 6.1](#). Use the truncation $N(m)$ of [proposition 9.1](#). We must check that the finite multiplier

$$S_m(s) = \sum_{r=0}^{N(m)-1} (-1)^r c_r 2^{2rs+2r^2+r} h^{2r}$$

remains nonzero near s_0 even though its number of terms grows. For $1 \leq r < N(m)$, the bound on the logarithm of the r th term is

$$2Lr^2 + \{L(1 + 2b - 2m) - \log R\}r + \log B.$$

This convex quadratic decreases until a point $m/2 + O(1)$. On the integer interval from 1 to $N(m) - 1$ its maximum is at an endpoint. The first endpoint is $-2Lm + O(1)$, while the last is $-Lm^2/2 + O(m)$. Thus $\sup_{s \in \mathcal{K}} |S_m(s) - 1| = O(m2^{-2m}) \rightarrow 0$. In particular $|S_m| \geq 1/2$ for large m .

The local lower bound $|C(s)| \geq g_0 |s - s_0|^d$ and [equation \(9.3\)](#) now allow Rouché on a circle of radius $\exp[-Lm^2/(2d) + B_1m]$, for a sufficiently large fixed B_1 . The comparison term strictly dominates the remainder. This gives the asserted d zeros. Identity [equation \(6.2\)](#) transfers the statement to $a = 1$, since these disks are eventually disjoint from the imaginary axis. $\square$

The proof is deliberately insensitive to the multiplicity of the limiting zero. For a simple zero the coefficient $1/d$ is absent; for a multiple zero it is the natural loss produced by a local factor $(s - s_0)^d$.

10 A non-lognormal measure with exact lognormal moments

10.1 Moment equality and genuine distinctness

Let V be lognormal with

$$\log V \sim N(L/2, L). \quad (10.1)$$

Its moments are $\mathbb{E}V^s = \exp[Cs(s+1)/2]$ for real s , and indeed for complex s . By [equation \(7.9\)](#),

$$\mathbb{E}T^n = \mathbb{E}V^n \quad (n \in \mathbb{Z}). \quad (10.2)$$

In logarithmic coordinates, the two densities are

$$f_{\log T}(v) = \frac{1}{C_0} e^{-v^2/(2L)+v/2} H(v), \quad (10.3)$$

$$f_{\log V}(v) = \frac{1}{\sqrt{2\pi L}} e^{-(v-L/2)^2/(2L)}. \quad (10.4)$$

Their density ratio is

$$\frac{d\mathbb{P}_T}{d\mathbb{P}_V}(e^v) = \frac{\sqrt{2\pi L} e^{L/8}}{C_0} H(v). \quad (10.5)$$

It is bounded above and below by positive constants, but it is not constant. Thus the measures are distinct, despite matching every positive and negative integer moment.

This is an explicit realization of a classical indeterminate moment problem. The new role of this particular representing measure is that it is built from the Thue–Morse product and supplies the scaling limit for the Dirichlet zero problem. The abstract fact that lognormal moments do not determine a measure is not new [7, Section 9].

10.2 Size bias and the periodic degree of freedom

The size-biased law of a positive random variable T with finite mean is defined by

$$\mathbb{E}f(T^*) = \frac{\mathbb{E}[Tf(T)]}{\mathbb{E}T}$$

for bounded measurable f . In the present normalization, $\mathbb{E}T = 2$. A change of variables and [equation \(2.9\)](#) give

$$T^* \stackrel{d}{=} 2T. \quad (10.6)$$

Indeed, the density of T^* is $K(t)/(2C_0)$, and the density of $2T$ is $K(t/2)/(C_0t) = K(t)/(2C_0)$.

More generally, let p be a bounded L -periodic measurable function. The same Mellin substitution used for [equation \(3.1\)](#) shows that

$$\int_0^\infty t^n p(\log t) \mu(dt) = 2^{n(n+1)/2} \int_0^\infty p(\log t) \mu(dt) \quad (n \in \mathbb{Z}). \quad (10.7)$$

If p has mean zero under μ , then $(1 + \theta p(\log t))\mu(dt)$ is a probability measure with the same integer moments whenever $|\theta| \|p\|_\infty \leq 1$. Choosing a nonconstant p produces infinitely many such measures.

To prove [equation \(10.7\)](#) for all integer n , define $C_p(s) = \int t^{s-1} K(t) p(\log t) dt$. Boundedness of p permits every integral, and its periodicity gives $C_p(s+1) = 2^{s+1} C_p(s)$ by the same substitution. Iteration at integer arguments proves the formula, including negative indices.

10.3 An exact mixture over geometric lattices

Write every $t > 0$ uniquely as $t = v2^k$ with $1 \leq v < 2$ and $k \in \mathbb{Z}$. Define

$$\Theta(v) = \sum_{k \in \mathbb{Z}} v^{-k} 2^{-k(k-1)/2}. \quad (10.8)$$

The series converges uniformly for $1 \leq v \leq 2$. Then

$$\mu = \int_1^2 \nu_v \frac{K(v)\Theta(v)}{C_0} \frac{dv}{v}, \quad (10.9)$$

where the discrete probability measure ν_v is supported on $\{v2^k : k \in \mathbb{Z}\}$ and has weights

$$\nu_v(\{v2^k\}) = \frac{v^{-k} 2^{-k(k-1)/2}}{\Theta(v)}. \quad (10.10)$$

To verify the identity, split the integral defining μ into the intervals $[2^k, 2^{k+1})$, substitute $t = 2^k v$, and apply [equation \(2.10\)](#). This also proves that the mixing density integrates to one.

Every single ν_v has the same integer moments:

$$\int t^n \nu_v(dt) = 2^{n(n+1)/2} \quad (n \in \mathbb{Z}). \quad (10.11)$$

In fact, shift the summation index by n in the numerator obtained from [equation \(10.10\)](#); completing the exponent gives the factor $v^{-n} 2^{n(n+1)/2}$, which cancels the external v^n . These geometric-lattice moment representatives are classical Chihara–Leipnik-type laws; see [7]. Formula [equation \(10.9\)](#) identifies the particular phase mixture selected by the Thue–Morse completion.

11 Exact moment algebra: determinants and orthogonal polynomials

The integer moments in [equation \(7.9\)](#) make a small amount of exact algebra available essentially for free. It provides useful checks on both the normalization and the relation to the Stieltjes–Wigert moment problem. The algebraic identities themselves are standard consequences of geometric moments, not independent priority claims.

Proposition 11.1 (Hankel determinant). *For every integer $d \geq 1$,*

$$\det \left[2^{(i+j)(i+j+1)/2} \right]_{i,j=0}^{d-1} = 2^{d^2(d-1)/2} \prod_{k=1}^{d-1} (2^k - 1)^{d-k}. \quad (11.1)$$

Proof. Factor $2^{i(i+1)/2}$ from row i and $2^{j(j+1)/2}$ from column j . The remaining matrix is $[(2^i)^j]$, whose determinant is the Vandermonde product $\prod_{0 \leq i < j < d} (2^j - 2^i)$. In this product factor 2^i from each difference. The powers of 2 from the rows and columns have total exponent $d(d-1)(d+1)/3$; the extra exponent is $d(d-1)(d-2)/6$. Their sum is $d^2(d-1)/2$. A difference $j-i=k$ occurs $d-k$ times, giving the remaining product. $\square$

Define the Gaussian binomial coefficient at $q = 2$ by

$$\begin{bmatrix} n \\ k \end{bmatrix}_2 = \prod_{j=0}^{k-1} \frac{2^{n-j} - 1}{2^{j+1} - 1} \quad (0 \leq k \leq n). \quad (11.2)$$

Empty products equal one.

Proposition 11.2 (Monic orthogonal polynomials). *The monic degree- n polynomial orthogonal to $1, t, \dots, t^{n-1}$ under μ is*

$$p_n(t) = \sum_{k=0}^n (-1)^{n-k} 2^{n(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_2 t^k. \quad (11.3)$$

Its squared norm is

$$\int p_n(t)^2 \mu(dt) = 2^{n(3n+1)/2} \prod_{k=1}^n (2^k - 1). \quad (11.4)$$

The same polynomials and norms apply to every other representing measure discussed in [section 10](#).

Proof. For a polynomial $p(t) = \sum_{k=0}^n a_k t^k$,

$$\int t^j p(t) \mu(dt) = 2^{j(j+1)/2} \sum_{k=0}^n a_k 2^{k(k+1)/2} (2^j)^k.$$

Thus orthogonality for $j = 0, \dots, n-1$ is equivalent to the polynomial identity

$$\sum_{k=0}^n a_k 2^{k(k+1)/2} z^k = 2^{n(n+1)/2} \prod_{j=0}^{n-1} (z - 2^j).$$

Expanding the product by the finite q -binomial identity gives [equation \(11.3\)](#). This proves both orthogonality and monicity. Strict positivity of the Hankel matrices guarantees uniqueness. The norm is the ratio of the size- $(n+1)$ and size- n Hankel determinants; substitution of [equation \(11.1\)](#) gives [equation \(11.4\)](#). Alternatively, the same ratio follows by triangular Gram–Schmidt elimination. Every integral used here is a polynomial moment, so it is unchanged for any representing measure with those moments. $\square$

For example,

$$p_0(t) = 1, \quad p_1(t) = t - 2, \quad p_2(t) = t^2 - 12t + 16.$$

The norm of p_1 is $8 - 4 = 4$, agreeing with [equation \(11.4\)](#). These low-degree checks are sensitive to the easily confused alternatives $2^{n(n-1)/2}$ and $2^{n(n+1)/2}$ in the moment normalization.

12 Identical asymptotic series, different positive functions

The moment interpretation gives a concrete nonuniqueness result for the all-orders expansion itself. Let V be the lognormal variable in [equation \(10.1\)](#), independent of the same W , and set

$$\tilde{\mathcal{H}}(x) = \mathbb{E}e^{-xV^2W}. \quad (12.1)$$

Theorem 12.1 (A completely monotone beyond-all-orders ambiguity). *The functions $\mathcal{H}$ and $\tilde{\mathcal{H}}$ are distinct, completely monotone functions with identical derivatives of every order at zero. Both satisfy the same alternating bound [equation \(7.11\)](#). Consequently, for every $N \geq 0$,*

$$\left| \mathcal{H}(x) - \tilde{\mathcal{H}}(x) \right| \leq 2A_N x^N \quad (x \geq 0). \quad (12.2)$$

Optimizing N as in [corollary 8.2](#) gives a logarithmic-Gaussian upper bound for their difference at small x .

Proof. Equality of the integer moments of T and V gives

$$\frac{\mathbb{E}(T^2W)^r}{r!} = \frac{\mathbb{E}(V^2W)^r}{r!} = A_r$$

for every integer $r \geq 0$. The proof of [theorem 7.2](#) applies unchanged to both Laplace transforms, so the derivatives and remainder bounds agree. Subtracting the same finite polynomial gives [equation \(12.2\)](#).

Suppose nevertheless that the two functions were identical. Uniqueness of Laplace transforms for probability measures on $[0, \infty)$ would imply $T^2W \stackrel{d}{=} V^2W$. For every real $r > 0$, all moments involved are finite and $\mathbb{E}W^r > 0$, so independence permits cancellation:

$$\mathbb{E}T^{2r} \mathbb{E}W^r = \mathbb{E}V^{2r} \mathbb{E}W^r \implies Q(2r) = 1.$$

The identity theorem would then force $Q \equiv 1$, contradicting [proposition 3.2](#). Thus the functions are distinct. $\square$

The factor 2 in [equation \(12.2\)](#) is convenient rather than optimal: since both functions lie in the same interval between successive alternating truncations, one can in fact replace it by 1. Keeping the weaker bound makes clear that even the triangle inequality already proves flatness. Either version is enough to show that every coefficient of the asymptotic expansion can be known exactly while the function still is not uniquely specified.

This phenomenon is not an arbitrary addition of a small oscillatory term. Both realizations are positive Laplace transforms and satisfy all complete-monotonicity inequalities. Thus positivity, smoothness, and a full asymptotic series still leave a nontrivial ambiguity. The Thue–Morse product supplies the extra information selecting $\mathcal{H}$ rather than its lognormal replacement.

13 Extension to sparse digit sequences in other integer bases

The completion is not restricted to base two. This section records the extension with enough detail to identify exactly where integrality of the base is used.

For a real number $b > 1$, define

$$P_b(t) = \prod_{j \geq 0} (1 - e^{-b^j t}), \quad X_b = \sum_{j \geq 1} b^{-j} U_j, \quad \Phi_b(t) = \mathbb{E} e^{-t X_b}, \quad K_b = P_b \Phi_b. \quad (13.1)$$

Now X_b is supported on $[0, 1/(b-1)]$ and has mean $a_b = 1/[2(b-1)]$. The same cancellation gives

$$K_b(bt) = \frac{K_b(t)}{t}, \quad C_b(s+1) = b^{s+1} C_b(s), \quad (13.2)$$

where $C_b(s) = \int_0^\infty t^{s-1} K_b(t) dt$. Thus the normalized measure has moments $b^{n(n+1)/2}$ for every integer n , and the exact logarithmic profile is periodic with period $\log b$.

When $b \geq 2$ is an integer, define $\mathcal{S}_b$ to be the nonnegative integers whose base- b digits are all zero or one. For $n \in \mathcal{S}_b$, let $w_b(n)$ be the number of nonzero digits. The product expansion gives

$$P_b(t) = \sum_{n \in \mathcal{S}_b} (-1)^{w_b(n)} e^{-nt}, \quad (13.3)$$

and hence the sparse signed Dirichlet series

$$D_b(s, a) = \sum_{n \in \mathcal{S}_b} \frac{(-1)^{w_b(n)}}{(n+a)^s}. \quad (13.4)$$

Absolute convergence holds at least in $\operatorname{Re} s > \log_b 2$: the integers in $\mathcal{S}_b$ below b^k are exactly 2^k in number, and comparison by base- b blocks proves convergence there. The Mellin integral continues the series to an entire function just as before.

Theorem 13.1 (Integer-base zero strings). *For every integer $b \geq 2$ and every $a > 0$, $D_b(\cdot, a)$ has infinitely many nonreal zeros in the left half-plane. More precisely, if s_0 is a zero of C_b of multiplicity d , the corresponding clusters approach $s_0 - m$ with error smaller than b^{-Am} for every fixed $A > 0$. At the centered shift $a = a_b$, their error is bounded by*

$$\exp \left[-\frac{\log b}{2d} m^2 + O(m) \right]. \quad (13.5)$$

Proof. All the completion and Mellin estimates replace L by $\log b$. The nonconstancy proof is still valid: because b^j is an integer, $P_b(x + 2\pi i) = P_b(x) \rightarrow 0$, whereas a constant-profile elementary Gaussian expression would have a nonzero limit there. The normalized periodic Mellin function is therefore nonconstant, entire, positive on the real axis, and of order at most two. Lemma 3.3 supplies a nonreal zero.

Put $B_{b,a}(t) = e^{-at}/\Phi_b(t)$. Since X_b has positive probability in every neighborhood of zero, the proof of the global Taylor remainder in lemma 5.1 applies. With $h = b^{-m}$, exact rescaling gives

$$b^{-m(m-1)/2+ms} \Gamma(s-m) D_b(s-m, a) = \int_0^\infty u^{s-1} K_b(u) B_{b,a}(hu) du.$$

At poles of the Gamma factor, the left side is interpreted through its entire Mellin numerator. The j th coefficient is again a multiple of $C_b(s+j)$, and equation (13.2) factors out $C_b(s)$. The proof of theorem 6.1 now applies verbatim.

For the centered bound,

$$B_{b,a_b}(z) = \prod_{j \geq 1} \frac{z/(2b^j)}{\sinh(z/(2b^j))}$$

is the Laplace transform in z^2 of

$$W_b = \sum_{j,n \geq 1} \frac{E_{j,n}}{4\pi^2 b^{2j} n^2}.$$

Its first moment-generating pole is at $R_b = 4\pi^2 b^2$. Removing the first pole as in [theorem 8.1](#) bounds its coefficients by a constant times R_b^{-r} . The counterpart of [equation \(9.2\)](#) then has exponent $2(\log b)N^2 - 2m(\log b)N + O(N)$. Optimizing at $N = m/2 + O(1)$ and repeating the Rouché argument proves [equation \(13.5\)](#). $\square$

For noninteger real $b > 1$, the positive product, Mellin identity, and moment construction still hold. The boundary-periodicity step used to prove nonconstancy does not automatically survive, and [equation \(13.3\)](#) is no longer an ordinary integer-indexed Dirichlet series. We do not silently extend the integer-base zero claim to that setting.

14 An explicit Fourier formula for the logarithmic profile

The proofs above need neither zeta values nor an evaluated constant. For reproducible computation, however, it is useful to make the profile explicit. Mellin analysis of harmonic sums is a standard method [\[8\]](#); the calculation is included here to fix every sign and normalization.

Let γ denote Euler’s constant and γ_1 the first Stieltjes constant in

$$\zeta(1+z) = \frac{1}{z} + \gamma - \gamma_1 z + O(z^2). \quad (14.1)$$

Define

$$\chi_k = \frac{2\pi i k}{L}, \quad c_* = -\frac{L}{12} - \frac{\pi^2/12 - \gamma^2/2 - \gamma_1}{L}. \quad (14.2)$$

Proposition 14.1 (Exact profile formula). *For real v ,*

$$\log H(v) = c_* - \frac{1}{L} \sum_{k \in \mathbb{Z} \setminus \{0\}} \Gamma(\chi_k) \zeta(1 + \chi_k) e^{-\chi_k v}. \quad (14.3)$$

The Fourier series and every fixed derivative converge absolutely.

Proof. For $\operatorname{Re} z > 0$, expanding $\log(1 - e^{-t})$ into exponentials and integrating termwise gives

$$\int_0^\infty t^{z-1} \log(1 - e^{-t}) dt = -\Gamma(z) \zeta(z+1).$$

Summing the dyadic dilates, also absolutely for $\operatorname{Re} z > 0$, gives

$$\log P(t) = -\frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \frac{\Gamma(z) \zeta(z+1)}{1 - 2^{-z}} t^{-z} dz, \quad c > 0. \quad (14.4)$$

This Mellin inversion is justified by exponential decay of the Gamma factor on vertical lines.

Move the line to $\operatorname{Re} z = -\delta$, with $0 < \delta < 1$. The crossed poles are $z = 0$ and $z = \chi_k$, $k \neq 0$. One may use horizontal sides at heights $2\pi(N + 1/2)/L$, which stay away from the zeros of the denominator. Stirling’s estimate for Γ , polynomial bounds for ζ on this fixed vertical strip, and the bound on the denominator show that the horizontal integrals tend to zero. On the new vertical line the integral is $O(t^\delta)$ as $t \downarrow 0$. The residue sum converges absolutely because $|\Gamma(\chi_k)|$ decays exponentially in $|k|$.

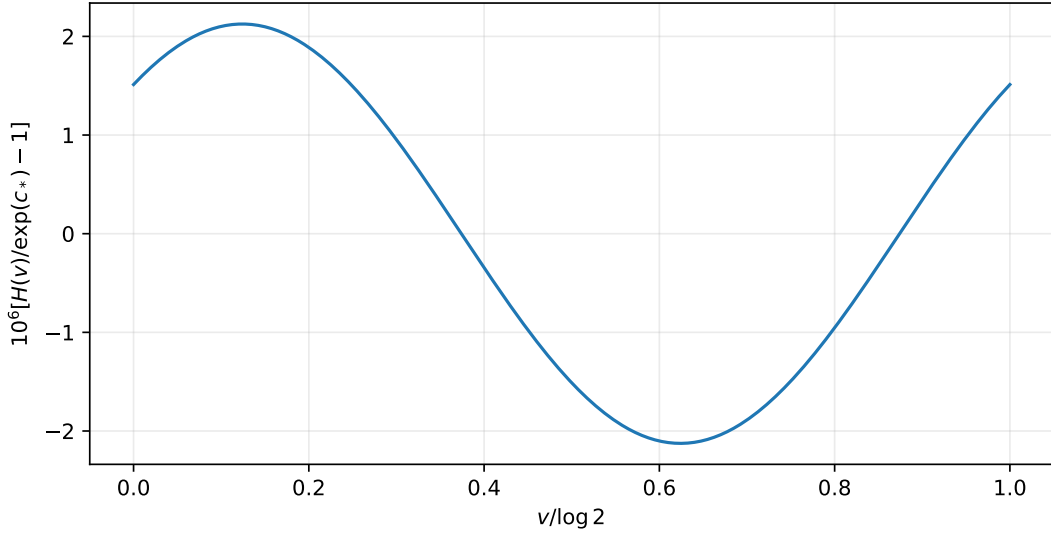


Figure 2: The nonconstant profile H over one logarithmic period. The vertical scale is in parts per million relative to e^{c_*} . This small oscillation is not a negligible detail for the complex zero problem.

At a nonzero χ_k , the derivative of $1 - 2^{-z}$ is L , so its residue contributes $-\Gamma(\chi_k)\zeta(1 + \chi_k)t^{-\chi_k}/L$. At zero, use

$$\begin{aligned}\Gamma(z)\zeta(1 + z) &= \frac{1}{z^2} + \left(\frac{\pi^2}{12} - \frac{\gamma^2}{2} - \gamma_1\right) + O(z), \\ \frac{1}{1 - e^{-Lz}} &= \frac{1}{Lz} + \frac{1}{2} + \frac{Lz}{12} + O(z^3).\end{aligned}$$

Multiplication by t^{-z} gives the contribution

$$-\frac{(\log t)^2}{2L} + \frac{\log t}{2} + c_*.$$

Consequently,

$$\log P(t) = -\frac{(\log t)^2}{2L} + \frac{\log t}{2} + c_* - \frac{1}{L} \sum_{k \neq 0} \Gamma(\chi_k)\zeta(1 + \chi_k)t^{-\chi_k} + O(t^\delta).$$

Since $\log \Phi(t) = O(t)$, comparison with the definition of H gives the claimed Fourier expression plus $O(t^\delta)$ for small t . Finally replace t by $2^{-m}t$. Both the profile and the Fourier sum are invariant under this substitution, while the error tends to zero. This turns the asymptotic identity into the exact formula [equation \(14.3\)](#). Exponential Gamma decay absorbs any fixed derivative multiplier χ_k^r . $\square$

The Fourier series for $\log H$ and the Fourier series for H should not be confused. To use [equation \(3.7\)](#), first exponentiate [equation \(14.3\)](#), then calculate the coefficients h_k of the resulting positive function. Substituting the coefficients of $\log H$ directly into [equation \(3.7\)](#) gives an incorrect normalization and shifts the computed zeros.

15 Numerical experiments and reproducibility

15.1 What was computed

The accompanying script code `verify.py` performs exact rational coefficient and determinant checks, evaluates K by its products, evaluates the profile by [equation \(14.3\)](#), and compares direct

Mellin quadrature with the Fourier–Gaussian representation. It also solves for selected zeros in the normalized variable, using [equation \(5.4\)](#) rather than a very large unnormalized Gamma factor. The files under `data/` contain the numerical outputs. The completion constant and the displayed limiting zero agree in separate 60- and 80-digit runs. The script `code/check_original_mellin.py` additionally tests the first listed Dirichlet zero directly in the original integral [equation \(4.1\)](#), for both $a = 1/2$ and $a = 1$, without using the completion to construct its integrand.

These are high-precision floating-point experiments. Their reported residuals are not interval-arithmetic error certificates. The existence theorems and the signed remainder bounds are proved analytically above and do not depend on interpreting a small floating-point residual as a proof.

The normalizing constant is approximately

$$C_0 = 0.7507219804078709592097079140506346947658 \dots \quad (15.1)$$

One numerically observed simple zero of C , chosen in a real fundamental strip, is

$$s_0 \approx 0.1240623273169621334141320413937 + 6.7214865334392714744741792585671 i. \quad (15.2)$$

The word “one” is intentional: the computation does not establish that this is the lowest-height nonreal zero, nor that all zeros are simple. The normalized derivative $Q'(s_0)$ is numerically extremely close to $2\pi i$, consistent with a leading two-term Fourier balance, but that observation is not used in the proof.

m	$\operatorname{Re} s_m$	$\operatorname{Im} s_m$	$ s_m - (s_0 - m) $
4	−3.8759680374867420	6.7214963407429050	$3.19093 \cdot 10^{-5}$
6	−5.8759376731062212	6.7214865336192914	$4.59882 \cdot 10^{-10}$
8	−7.8759376726830382	6.7214865334392717	$4.19307 \cdot 10^{-16}$

Table 1: Numerical zeros of $D(s, 1/2)$ and, by [equation \(6.2\)](#), of $D(s, 1)$. The distances refer to the higher-precision value of s_0 stored in the data file.

The centered slope ratios provide an independent positive-axis check:

m	$Z_m(1/2)/(C_0 2^{m(m-1)/2})$
0	0.9233074275823854568421676411
2	0.9934028797641805193051831301
4	0.9995677618356360314893011938
6	0.9999728805385416834067380219
8	0.9999983046075012694941809752

Table 2: Direct Mellin quadrature of the exact samples $\mathcal{H}(4^{-m})$. The script checks the signed inequalities for truncation orders one through seven.

15.2 Stable product evaluation

For positive t , compute $\log P(t)$ as a sum of $\log(-\expm1(-2^j t))$ rather than by subtracting nearly equal floating-point numbers. Its omitted large-scale terms decrease doubly exponentially.

For $\Phi(t)$, first apply enough halvings that the remaining argument v lies in $(0, 1]$. Multiply the finitely many removed factors exactly at working precision. The infinite remaining tail is then computed from

$$\log \Phi(v) = -\frac{v}{2} + \sum_{r \geq 1} \frac{B_{2r} v^{2r}}{2r(2r)!(2^{2r} - 1)}. \quad (15.3)$$

This avoids multiplying hundreds of factors close to one. Both the product route and the Fourier route are retained in the script so that they can cross-check one another.

For the zeta values in [equation \(14.3\)](#), the points $1 + 2\pi ik/L$ are exact dyadic resonances. An implementation using $\zeta(s) = \eta(s)/(1 - 2^{1-s})$ naively can lose precision there, since both numerator and denominator vanish. The supplied code explicitly requests Euler–Maclaurin evaluation of ζ . In the tested `mpmath` environment this avoided a visible loss of accuracy in the profile. This is an implementation detail, not a mathematical singularity of ζ at these points.

15.3 Why the Mellin quadrature is manageable

In logarithmic coordinates, the integrand for C is a Gaussian multiplied by a bounded periodic function and e^{sv} . For a fixed complex s , the center of its absolute Gaussian majorant is $L(\operatorname{Re} s + 1/2)$. The code chooses its finite integration interval using that center and the requested precision. At the centered shift, $0 < \Psi(2^{-m}e^v) \leq 1$, so the same Gaussian tail estimate also controls $F_{m,1/2}$.

The finite interval and quadrature discretization are not interval-certified in this implementation. Repeating at increased precision and comparing two analytic representations gives a strong consistency check, while the proofs supply the logical foundation. The exact rational calculations, by contrast, are exact: the coefficient formulas and Hankel determinants are evaluated with SymPy rationals and integers.

15.4 Reproduction commands

From the article directory, a typical run is

```
python -m pip install -r code/requirements.txt
python code/verify.py --dps 60 --roots --figures
python code/verify.py --dps 80
pdflatex -interaction=nonstopmode Thue_Morse_Dyadic_Completion.tex
pdflatex -interaction=nonstopmode Thue_Morse_Dyadic_Completion.tex
```

The PDF figures are included in the archive, so compiling the article does not require running Python first. The exact versions used for the stored computation are recorded in `code/requirements.txt`. No network requests are made by the verification script.

16 Relation to the atlas and a formalization route

16.1 Repository audit and limits of the novelty claim

The requested atlas source was inspected through the repository’s native file interface. The retrieved source blob had SHA

7be75f3bc86f9c6712eec6d7550eaea53716434a.

The document’s front matter identifies it as consolidated on 28 August 2026. The comparison concentrated on the Laplace-product, boundary-flatness, Mellin–Dirichlet, and master-product sections, with targeted repository searches for related moment and zero material. This was not a line-by-line audit of every file in the large surrounding research tree.

The atlas already establishes the positive product, effective flatness, entire continuation, zeros at nonpositive integers, the dyadic parameter equation, and the bridge from $D'(0, a)$ to Thue–Morse infinite products. These are inputs or short consequences in the present article, not asserted additions to the atlas. The surrounding corpus also contains spectral moment constructions related to Stieltjes–Wigert measures and reciprocal-transform polynomials. Accordingly, the existence of such general themes is not presented as novel either.

The specific extension developed here is the completion $P\Phi$ with its exact Mellin shift, its nonconstant entire periodic factor, the all-orders factorization of the translated Dirichlet family, and the resulting zero strings. The centered exponential-mixture representation then upgrades a formal coefficient calculation to a signed, globally valid error theorem and an optimized bound. Those are the statements whose proofs should be compared against future literature findings.

The public literature checked included the original Dirichlet-series work and later automatic-series framework [2, 3], Allouche’s zero question [4], Tóth’s related Dirichlet identities [5], Fabius-function treatments [9, 10], and the probability survey [7]. The relevant Alkauskas preprint was identifiable bibliographically but its full text was not accessible during this investigation. Absence of a matching result from these searches is not proof of priority.

16.2 A proof-oriented dependency chain

A formal development can be organized into four layers, without requiring the numerical Fourier formula for the principal theorem. First prove positivity and normal convergence of P and Φ , then their two scaling equations. This yields $K(2t) = K(t)/t$ and positive upper and lower bounds for its logarithmic profile.

Next establish the Gaussian domination needed for the entire Mellin transform and its derivatives. The Mellin shift, periodicity of Q , and the boundary contradiction proving nonconstancy can then be formalized independently of coefficient calculations. The zero-free finite-order lemma and a version of Rouché with multiplicities complete the complex-analytic layer.

The asymptotic layer consists of the positive-axis global Taylor bound for B_a and the exact change of variables in [equation \(5.4\)](#). Once these are available, the all-orders factorization is a finite-sum integration lemma, and the supergeometric zero-cluster theorem is a uniform remainder estimate followed by Rouché.

Finally, the centered layer adds Euler’s sinh product, the positive exponential-sum law W , and an integral Taylor remainder. It supplies the signed bounds, the coefficient-pole estimate, and the Gaussian refinement. The finite Bell-polynomial formula and the Hankel determinant can be developed in a separate exact algebra module.

The existing atlas names modules such as `ThueMorseBoundaryFlatness.lean` and `ThueMorseEntireContinuation.lean` for relevant background. No new Lean file is supplied here, and no theorem in this extension is claimed to have been machine-checked. The dependency chain is a technical implementation guide, not a formalization status claim.

17 Further questions suggested by the proved results

The following questions concern refinements not needed for any preceding theorem.

Question 17.1 (Zeros in a fundamental strip). Can the nonreal zeros of Q in $0 \leq \operatorname{Re} s < 1$ be counted and localized effectively as their imaginary parts grow? Are they all simple?

The Fourier–Gaussian representation is a natural starting point. At increasing height, different Fourier harmonics compete. A rigorous two-term dominance argument on suitable contours could produce explicit zero enclosures, but the present computation is not a proof of such a global pattern.

Question 17.2 (Leading exponentially small displacement). For a simple zero s_0 of C , determine an asymptotic equivalent, not just an upper bound, for $s_m - (s_0 - m)$ at the centered shift.

All algebraic powers of 2^{-m} vanish from the displacement. The nearest poles of Ψ and the logarithmic-Gaussian saddle suggest a more detailed exponentially small analysis. Such an analysis must retain the phase information in H ; replacing the completion by its lognormal twin would erase the zeros of the limiting periodic factor.

Question 17.3 (Growing-height zero windows). How far may $|\operatorname{Im} s|$ grow with m while the translated asymptotic expansion remains useful for zero localization?

The remainder bound in [equation \(5.6\)](#) depends on $\operatorname{Re} s$ but not directly on $\operatorname{Im} s$. The difficulty is instead a lower bound for $C(s)$ on the contours used by Rouché. Thus this question couples growth of the periodic entire factor to the geometry of its zeros.

Question 17.4 (Noninteger dilation). For every real $b > 1$, is the logarithmic profile of K_b nonconstant? What replaces the integer-frequency boundary argument when b is not an integer?

An affirmative answer would extend the Mellin zero mechanism to a continuous family of generalized exponential sums. The exact completion and moment identities already hold; the missing step in the proof given here is the nonconstancy argument in that greater generality.

Closing perspective

The most useful organizing fact is a cancellation between an expanding product and a contracting one. The Thue–Morse product alone carries a complicated dyadic scale dependence; the uniform-convolution transform removes its nonmonomial factor. The resulting exact law $K(2t) = K(t)/t$ simultaneously explains lognormal integer moments, the negative-argument scale of the Dirichlet continuation, the universal factor in every asymptotic coefficient, and the occurrence of nonreal zero strings. At the half-shift, symmetry supplies an additional positive Laplace representation and converts the expansion into certified inequalities. The log-periodic information is small on the real axis, but it is precisely the information that survives beyond all orders and controls the complex zeros.

A Coefficient register and normalization checks

The first six coefficients in the centered expansion are

r	A_r in $\mathcal{H}(x) \sim \sum_{r \geq 0} (-1)^r A_r x^r$
0	1
1	1/9
2	248/2025
3	4806656/2679075
4	3993751257088/10247461875
5	454890030220640780288/345944065438125

All are obtained by the finite formula [equation \(7.6\)](#) followed by multiplication by 2^{2r^2+r} . Their rapid growth is predicted by [theorem 8.1](#); it is not numerical instability.

The following identities are convenient independent checks:

$$\begin{aligned} \Phi(0) &= 1, & \Phi'(0) &= -\frac{1}{2}, & [z^2] \log \Phi(z) &= \frac{1}{72}, \\ K(2t)t &= K(t), & C(1) &= 2C_0, & C(-1) &= C_0, \\ \mathbb{E}T^2 &= 8, & \mathbb{E}W &= \frac{1}{72}, & \mathbb{E}(T^2W) &= \frac{1}{9}, \\ Z_m(1/2) &= C_0 2^{m(m-1)/2} \mathcal{H}(4^{-m}), \\ D(s, 1/2) &= (1 + 2^s)D(s, 1). \end{aligned}$$

There is also an exact normalization check at $m = 0$:

$$Z_0(1/2) = L, \quad Z_0(1) = L/2, \quad C_0 \mathcal{H}(1) = L.$$

Indeed, $P(2u) - P(u) = e^{-u}P(2u)$, so substitution $t = 2u$ gives

$$M(0, 1/2) = \int_0^\infty \frac{P(2u) - P(u)}{u} du = L.$$

To justify the last equality without splitting divergent integrals, integrate first over $[\epsilon, R]$. The difference is $\int_R^{2R} P(u) du/u - \int_\epsilon^{2\epsilon} P(u) du/u$; use $P(\infty) = 1$ and $P(0) = 0$. Differentiating [equation \(6.2\)](#) at zero gives the value at $a = 1$. These are classical product-normalization consequences, recorded here as checks rather than new evaluations.

A sign error in the logarithmic centering changes the odd coefficients; a missing factor of 2 in the Mellin scaling changes the moment law; and confusing $\Psi(z)$ with $\Psi(\sqrt{z})$ changes the optimal truncation scale. Each of these errors is detected by at least one of the checks above.

B A direct nonrecursive formula for every asymptotic coefficient

For readers wishing to implement the general-shift expansion without Bell-polynomial conventions, define $b = 1/2 - a$ and

$$d_r = -\frac{B_{2r}}{2r(2r)!(2^{2r} - 1)}.$$

Then its coefficient $\alpha_j(a) = 2^{j(j+1)/2} \beta_j(a)$ is

$$\alpha_j(a) = 2^{j(j+1)/2} \sum_{\substack{n_0, n_1, \dots, n_{\lfloor j/2 \rfloor} \geq 0 \\ n_0 + 2n_1 + 4n_2 + \dots + 2\lfloor j/2 \rfloor n_{\lfloor j/2 \rfloor} = j}} \frac{b^{n_0}}{n_0!} \prod_{r=1}^{\lfloor j/2 \rfloor} \frac{d_r^{n_r}}{n_r!}. \quad (\text{B.1})$$

The sum is finite. At $b = 0$, only even $j = 2r$ survive, and $\alpha_{2r}(1/2) = (-1)^r A_r$. For the complex translated expansion, simply multiply its j th coefficient by 2^{js} before multiplying by $C(s)$. Thus a single finite coefficient formula covers both the real-zero slopes and the complex zero-string expansion.

References

- [1] Vladimir Reshetnikov’s ProveIt repository, *The Thue–Morse Sequence: Formula Atlas and Fabius–Rvachev Frontier Results*, consolidated 28 August 2026; source inspected 4 September 2026.
Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/thue-morse/Thue_Morse_Atlas_and_Frontiers/Thue_Morse_Atlas_and_Frontiers.tex
https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/thue-morse/Thue_Morse_Atlas_and_Frontiers.
- [2] J.-P. Allouche and H. Cohen, *Dirichlet series and curious infinite products*, Bulletin of the London Mathematical Society **17** (1985), 531–538. <https://doi.org/10.1112/blms/17.6.531>.
- [3] J.-P. Allouche, M. Mendès France, and J. Peyrière, *Automatic Dirichlet series*, Journal of Number Theory **81** (2000), 359–373. <https://doi.org/10.1006/jnth.1999.2487>.
- [4] J.-P. Allouche, *Thue, Combinatorics on words, and conjectures inspired by the Thue–Morse sequence*, arXiv:1401.3727, version 2 (2014), Section 3.2. <https://arxiv.org/abs/1401.3727>.
- [5] L. Tóth, *Linear combinations of Dirichlet series associated with the Thue–Morse sequence*, Integers **22** (2022), A98. <https://arxiv.org/abs/2211.13570>.
- [6] G. Alkauskas, *Dirichlet series, associated with Thue–Morse sequence*, Vilnius University, Department of Mathematics, preprint 2001-15 (2001). Bibliographical reference only: the full text was not retrieved in this investigation. Author’s publication list: <https://web.vu.lt/mif/g.alkauskas/math/index.html>.
- [7] R. Arratia, L. Goldstein, and F. Kochman, *Size bias for one and all*, Probability Surveys **16** (2019), 1–61, especially Section 9. <https://doi.org/10.1214/13-PS221>; <https://arxiv.org/abs/1308.2729>.
- [8] P. Flajolet, X. Gourdon, and P. Dumas, *Mellin transforms and asymptotics: Harmonic sums*, Theoretical Computer Science **144** (1995), 3–58. [https://doi.org/10.1016/0304-3975\(95\)00002-E](https://doi.org/10.1016/0304-3975(95)00002-E).
- [9] J. Arias de Reyna, *Arithmetic of the Fabius function*, Integers **18** (2018), A51. <https://math.colgate.edu/~integers/s51/s51.pdf>.
- [10] J. K. Haugland, *Evaluating the Fabius function*, arXiv:1609.07999 (2016), version 2 (2020). <https://arxiv.org/abs/1609.07999>.